# ENSEMBLE LEARNING FOR SOLAR PANEL DEGRADATION DETECTION

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**Abstract**

The long-term performance of solar panels is often compromised by physical and environmental degradation such as cracks, hotspots, delamination, soiling, and discoloration. Timely detection of these defects is essential for maintaining energy efficiency and extending panel lifespan. This study proposes a deep learning–driven framework for automated multi-defect detection and classification in photovoltaic modules using advanced computer vision architectures, including YOLO variants, Region-Based Convolutional Neural Networks (R-CNNs), and Vision Transformers (ViT). The system leverages both RGB and thermal images to capture complementary visual and temperature-based information, enabling more accurate identification of multiple defect types. In addition, it quantifies defect severity through a computed degradation score reflecting the extent and impact of each anomaly. Experimental evaluations demonstrate that the proposed approach achieves high detection accuracy and provides a scalable solution for predictive maintenance, thereby enhancing the operational reliability of large-scale solar installations.

## I. Introduction

India's rapidly expanding solar energy capacity has positioned it among the global leaders in renewable energy adoption. However, environmental and structural challenges continue to threaten the operational efficiency and longevity of photovoltaic (PV) systems. Solar installations across the country, particularly in southern regions such as Tamil Nadu, are frequently exposed to extreme heat, dust accumulation, and monsoon-induced stress, which accelerate degradation and energy loss. A recent report by *The Times of India* highlighted how a routine storm exposed severe quality and maintenance issues in several solar farms, underscoring the urgent need for intelligent monitoring and automated inspection mechanisms [1]. Furthermore, regional studies, such as that conducted by Subha et al. [2], have identified multiple barriers to effective solar energy implementation in Tamil Nadu—ranging from inadequate technical maintenance frameworks to limited awareness of predictive inspection technologies.

Photovoltaic panels, which convert sunlight into electrical energy through semiconductor materials, are highly sensitive to environmental and operational stresses. Over time, several forms of degradation can develop, affecting both energy output and system longevity. Dust accumulation is one of the most common issues in tropical regions like India, where suspended particulates and pollution settle on panel surfaces, blocking sunlight and significantly reducing conversion efficiency if left uncleaned [6], [7]. Discoloration, often caused by prolonged ultraviolet (UV) exposure, moisture ingress, or encapsulant degradation, alters the optical properties of the panel surface, leading to uneven current generation and accelerated aging [18]. Cracks, typically arising from mechanical stress, thermal expansion, or hail impact, disrupt the electrical pathways between solar cells, resulting in partial shading, local heating, and accelerated cell failure [11].

Hotspots occur when certain cells generate excessive heat due to shading, micro-cracks, or manufacturing defects, potentially melting solder joints or damaging encapsulant layers [18]. Finally, Potential-Induced Degradation (PID) arises from high-voltage stress and ion migration between the glass surface and the semiconductor, causing significant leakage currents and long-term efficiency loss [8], [9].

Together, these defects not only diminish the power yield of solar arrays but also increase maintenance costs and reduce the overall lifespan of PV systems, underscoring the need for robust, automated inspection solutions [10]. To address these challenges, this paper proposes a deep learning–based framework that integrates architectures including YOLO variants, Masked Region-based Convolutional Neural Networks (Masked R-CNNs), and Vision Transformers (ViTs) for the automated detection and classification of solar panel defects. In addition, a degradation scoring model is introduced to quantify defect severity and predict maintenance requirements, facilitating proactive asset management and improving the reliability of large-scale solar power systems under diverse Indian climatic conditions [11].

The proposed architecture implements a dual-pipeline fault-analysis system for solar panels: RGB and thermal images are independently pre-processed and fed into dedicated image segmentation models. Each model is tailored to detect specific defect types: cracks, dust accumulation, and discoloration from RGB images, and potential-induced degradation (PID) and hotspots from thermal images. For every detected fault, a "defect measure" is computed, which is subsequently weighted according to the fault type, detection confidence, and affected area. The overall degradation score for a panel is then derived by summing the product of the defect measure and impact weight across all detected faults, providing a comprehensive assessment of panel health.

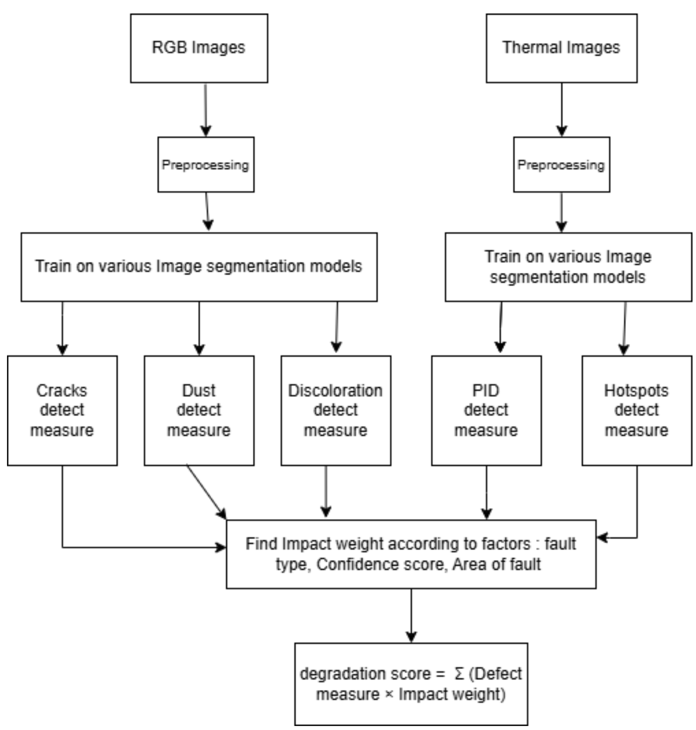



Fig. 1: Dual-pipeline architecture for automated solar panel defect detection.

## II. Literature Review

- Overview of existing studies in solar panel defect detection using image processing and machine learning.
- Summary of methods using CNNs, R-CNNs, YOLO, and Transformer architectures.
- Comparison of techniques: accuracy, speed, data requirements, and limitations.
- Identified research gaps that motivate this work.

## III. Methodology

### A. *Dataset and Preprocessing*

This study utilizes two complementary datasets to detect and classify visual and thermal defects in photovoltaic (PV) modules: (i) an RGB image dataset for surface-level defect analysis, and (ii) a thermal infrared image dataset for detecting temperature-related anomalies.

*1) RGB Dataset and Preprocessing:* The RGB dataset was obtained from the publicly available Kaggle repository "Solar Panel Images" [3]. It comprises a total of 556 images distributed across four categories: Dust (190), Cracks (69), Discoloration (103), and Clean panels (194), captured under varying environmental and lighting conditions. Clean panel images were included across all training subsets to improve defect discrimination and reduce class imbalance. Distinct preprocessing and augmentation pipelines were applied to each defect category to optimize feature extraction and enhance model generalization.

*a) Dust Dataset:* Preprocessing included auto-orientation correction and resizing with stretching to 640×640 pixels. Data augmentation generated three samples per training image, incorporating horizontal and vertical flips, zoom-based cropping (7%–22%), and brightness and exposure adjustments ranging from –19% to +19% and –7% to +7%, respectively. This resulted in a total of 368 images after augmentation.

*b) Discoloration Dataset:* Preprocessing involved auto-orientation, resizing to fit within 640×640 pixels, contrast stretching for auto-adjustment, and grayscale conversion to highlight tone-based anomalies. Three augmented samples were created per image, including horizontal and vertical flips, rotations at 90°, 180°, and 270°, random angular rotations between –15° and +15°, brightness variation between –25% and +25%, and random noise affecting up to 0.16% of pixels, resulting in 373 total images.

*c) Crack Dataset:* Images underwent auto-orientation, resizing with black-edge fitting to 640×640, and contrast enhancement via adaptive histogram equalization to emphasize micro-crack features. Augmentation generated three outputs per input image, with transformations including horizontal and vertical flips, angular rotations between –15° and +15°, brightness variation between –17% and +17%, and random pixel noise up to 0.81%. In total, 605 augmented images were produced. Across all RGB subsets, preprocessing preserved crucial texture and contrast features, while augmentation improved robustness to illumination and orientation variability.

(a) Solar Panel with Dust (b) Solar Panel with Discoloration (c) Solar Panel with Cracks

Fig. 2: RGB Images with defects

*2) Thermal Dataset and Preprocessing:* The thermal dataset was obtained from the Kaggle repository "Solar Panel Thermal Images" [4] It consists of 2,723 original thermal infrared images annotated with 7,772 labeled objects across eight YOLOv9 classes, representing a diverse set of thermal anomalies: Single Hotspot, Multi Hotspots, Single Diode, Multi Diode, Single Bypassed Substring, Multi Bypassed Substring, String (Open Circuit), and String (Reversed Polarity).

Preprocessing included EXIF-based orientation correction, resizing to 640×640 pixels, and grayscale normalization to ensure consistent thermal intensity scaling across samples. To enhance model generalization and improve representation of thermal behaviors, data augmentation generated approximately 7,500 additional samples using flip, rotation, shear, brightness, hue, and exposure variations.

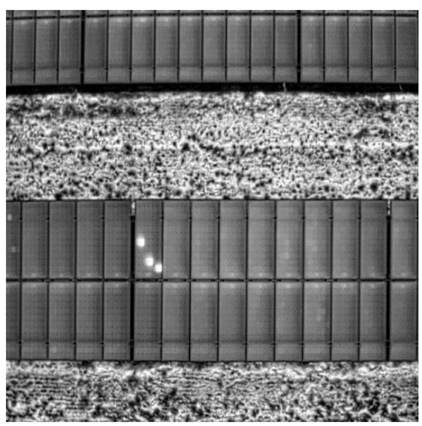


Fig. 3: Thermal Image with Hotspot

### *B. Model Architectures*

Following dataset preparation and preprocessing, the analytical pipeline employs a modular, multi-model architecture tailored to the diverse physical characteristics of defects observed in photovoltaic (PV) modules. Each computational component is selected to align with the specific visual or thermal signature of its target defect category, ensuring methodological coherence with real-world degradation mechanisms. Crack detection and segmentation are performed using a YOLOv11 instance segmentation model, optimized for fine structural defects in RGB imagery. Discoloration is quantified using a Roboflow instance segmentation framework capable of capturing diffuse region-based degradation patterns. Dust deposition is assessed through a Vision Transformer–based classifier that leverages self-attention mechanisms to capture global texture cues. Thermal anomalies and hotspot patterns are localized via a YOLOv8 detector trained on infrared imagery, while Potential Induced Degradation (PID) severity is estimated using a temperature-aware and topology-aware model integrating $\Delta T$ differentials with spatial clustering. The following subsections provide detailed descriptions of the architecture, training methodologies, and severity computation strategies implemented for each defect-analysis module within the integrated system.

*1) **Crack Detection using YOLOv11 Instance Segmentation**:* Crack detection in RGB imagery was implemented using a YOLOv11-based instance segmentation framework. YOLOv11 offers a unified, single-stage architecture capable of jointly performing object localization and pixel-level segmentation, making it well-suited for real-time defect analysis in photovoltaic (PV) modules [11]–[13]. Given an input RGB image, the model simultaneously predicts bounding boxes, segmentation masks, and class confidence scores corresponding to crack regions, thereby enabling a detailed structural characterization of each defect [14], [15].

Model development was conducted using a curated dataset of solar panel images annotated on the Roboflow platform [16]. Roboflow facilitated consistent polygon-based annotations, which were particularly advantageous for capturing the irregular geometries and fine-grained structures of micro-cracks. In addition, Roboflow's preprocessing pipeline ensured uniformity across the dataset through controlled transformations and mask generation. The segmentation head within the YOLOv11 architecture played a critical role by delivering high-resolution crack masks, which are essential for subsequent quantitative severity assessment [19].

Training of the YOLOv11 model employed the Stochastic Gradient Descent (SGD) optimizer with momentum, providing stable convergence across epochs and effective exploration of the model's parameter space [13]. The composite loss function integrated bounding box regression, classification, and segmentation losses, enabling the network to jointly optimize localization precision and mask accuracy. Training progressed until the validation loss reached a stable plateau, indicating that the model had achieved optimal feature extraction without overfitting [19].

**Unified Severity Scoring Model:** To evaluate the functional impact of detected cracks, a severity scoring methodology was designed. The final severity score integrates three principal physical and structural factors:

- Crack Area ($A_c$): The segmented crack area is quantified relative to the total panel area. Larger crack regions correspond to greater reductions in active photovoltaic surface area and hence higher severity [17], [18].
- Number of Cracks ($N_c$): The presence of multiple cracks indicates distributed mechanical stress, progressive material fatigue, or extensive cell-level degradation, contributing to a higher severity classification [17].
- Crack Location ($L_c$): Defects situated near interconnects, busbars, or cell boundaries are assigned greater significance due to their higher potential to disrupt current flow paths. In contrast, micro-cracks confined to surface regions are assigned lower weights, while edge-localized cracks receive intermediate weights [17].

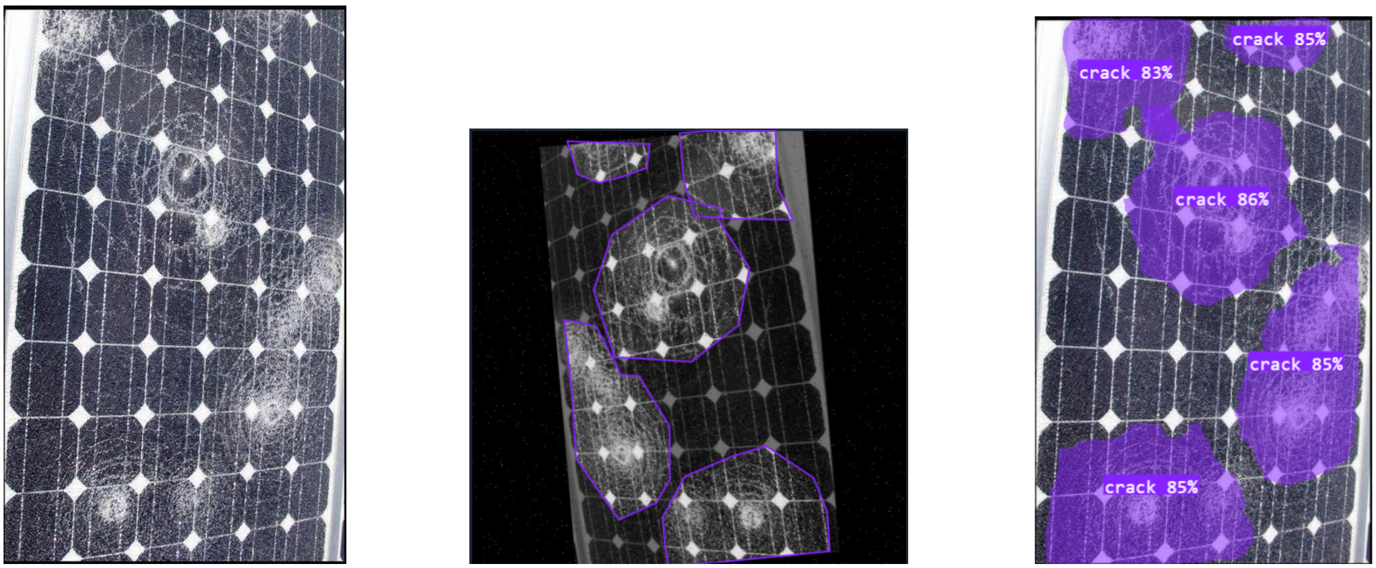
The severity score is computed as:

$$S_c = a_1 \cdot A_c + a_2 \cdot N_c + a_3 \cdot L_c, \tag{1}$$

where $(a_1, a_2, a_3)$ are equal coefficients (0.33) derived from state-level PV fault-criticality guidelines [8].

The segmentation mask predictions from YOLOv11 enable precise estimation of the crack area, supporting accurate severity analysis [11].

Model performance was assessed using standard object detection and segmentation metrics, including mean Average Precision at an IoU threshold of 0.5 (mAP@0.5), AP75, precision, and recall [19]. Comparative evaluations demonstrated that the YOLOv11 instance segmentation model outperformed baseline architectures such as Roboflow Instance Segmentation 3.0 and RCNN, achieving superior accuracy in both crack localization and mask quality [14], [15].




(a) Crack defect (b) Crack preprocessing (c) Crack prediction

Fig. 4: Crack detection pipeline

*2) **Discoloration Detection using Roboflow Instance Segmentation**:* Discoloration detection was implemented using the Roboflow Instance Segmentation 3.0 model, which provides a unified framework for simultaneous extraction of spatial location and pixel-level distributions of discoloration patterns in RGB solar panel imagery. This architecture is particularly advantageous for capturing diffuse degradation characteristics—such as browning, yellowing, hazing, and non-uniform color fading—that typically lack the sharp structural boundaries observed in crack defects [17], [18], [22].

The model produces bounding boxes, segmentation masks, confidence scores, and region contours, thereby enabling detailed identification and quantification of discolored regions [21]. Because discoloration manifests as soft, region-based optical degradation with gradual intensity transitions, segmentation-based analysis offers superior fidelity compared to bounding-box detection alone [14], [15]. The pixel-level masks produced by the Roboflow model allow precise estimation of affected area and spatial density, both of which are essential for assessing the operational impact of discoloration on photovoltaic performance [11], [19].

Training data were prepared and annotated using the Roboflow platform [16], which ensured consistent dataset organization and high-quality segmentation masks. Manual annotation of discolored regions captured variations in intensity, gradient patterns, and irregular spatial distributions, ranging from subtle yellowing to extensive dark-brown patches [6], [7]. The Roboflow preprocessing and augmentation pipeline—comprising geometric transformations, color normalization, and contrast adjustments—enhanced the model's ability to generalize across illumination conditions, camera variations, and different panel textures [12], [13], [20]. The model was trained using Roboflow's cloud-based optimization system [16], which automatically configures hyperparameters and performs continuous validation during training. The loss function integrated segmentation loss, bounding box regression loss, and classification loss to enforce balanced learning of region localization and mask accuracy [14], [15]. Adaptive learning-rate scheduling and convergence monitoring ensured stable performance throughout the training process, with training terminating when validation loss plateaued and mAP metrics stabilized [19], [22].

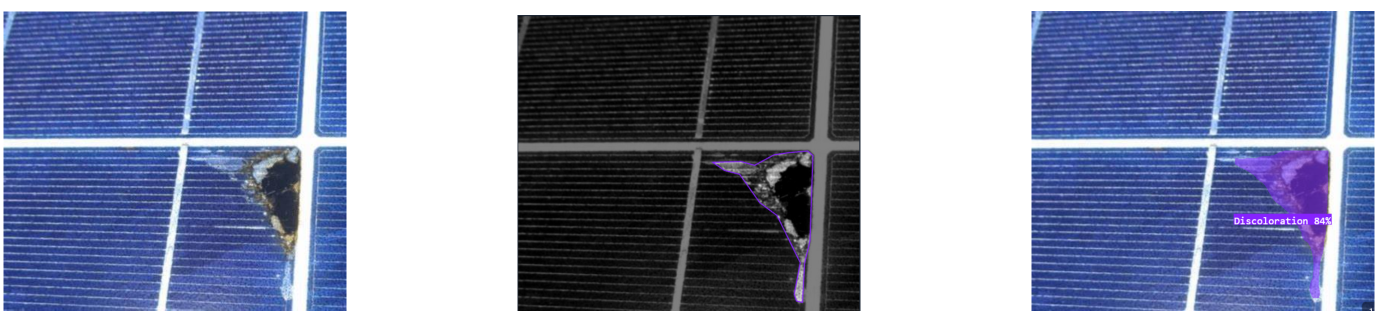
**Unified Severity Scoring Model:** To quantify the operational significance of discoloration, a severity scoring model was designed [21]. The final severity score incorporates the Discoloration Area, computed as the ratio of segmented discolored pixels to the total pixel area of the panel. Larger discolored regions correspond to reduced irradiance absorption and increased optical losses [17], [18].

$$\text{Discolored\_Area} = \sum_{i=1}^{N} \nVdash (M_i > 0)\,, \tag{2}$$

where $M_i$ denotes the $i$-th pixel of the predicted mask and $N$ is the total number of image pixels. The severity score is then defined as:

$$S_d = \frac{\text{Discolored\_Area}}{A_{\text{img}}} \times 100, \tag{3}$$

where $A_{\text{img}}$ represents the total image area in pixels.This formulation ensures that the severity measure reflects the spatial extent of discoloration, enabling a quantitative assessment of its impact on photovoltaic module performance [11], [19], [22].




(a) Discoloration defect (b) Discoloration preprocessing (c) Discoloration prediction

Fig. 5: Discoloration detection pipeline

*3) **Dust Classification using Vision Transformer (ViT):*** Dust detection was implemented using a Vision Transformer (ViT)–based classification model trained via the Roboflow cloud training platform [21], [23], [26]. Unlike structural defects such as cracks or discoloration, which appear as localized regions that can be segmented or detected through bounding boxes, dust accumulation generally spreads across the entire module surface in a diffuse and non-uniform manner [6], [7], [27]. In typical field conditions, dust does not form a distinct object or sharply bounded region, instead, it produces a global reduction in surface reflectance and a subtle alteration in texture patterns across the whole panel [28]. These characteristics make instance segmentation or object detection unsuitable, as such methods inherently assume well-defined spatial boundaries [7]. In contrast, a whole-image classification approach effectively captures the global visual changes associated with dust deposition, enabling robust differentiation between "dust-covered" and "clean" panels even when dust distribution is spatially irregular or nearly uniform [23]. The ViT architecture leverages multi-head self-attention mechanisms to capture long-range contextual relationships within the image [26]. This enables expressive modeling of subtle global texture variations and tonal irregularities that differentiate "dust-covered" surfaces from clean solar panel regions [28]. As a result, the ViT model is particularly effective in handling the homogeneous yet visually nuanced appearance of dust deposition [27].

The system produces a binary prediction ("dust" or "clean") accompanied by confidence scores for each class [21]. Because dust accumulation often manifests as broad, diffuse surface-level coverage rather than sharply bounded defects, whole-image classification provides an efficient and scalable detection strategy [6], [7].

Roboflow's ViT training pipeline was configured with the following components:

- **Optimizer:** AdamW, chosen for its adaptive learning-rate behavior and weight-decoupled regularization suited to transformer architectures [23].
- **Loss Function:** Categorical cross-entropy, ensuring stable gradient updates for the binary classification task [26].
- **Epoch Scheduling:** Early stopping based on validation-loss convergence to prevent overfitting and ensure optimal generalization [28].

**Unified Severity Scoring Model:** To convert classifier outputs into an operationally meaningful dust severity measurement, a custom severity scoring formulation was developed [27]. The model's confidence score for the predicted label serves as the basis for continuous severity estimation. Severity is defined as:

$$S_{\text{dust}} = \begin{cases} 100 \times \text{confidence}, & \text{if label = dust,} \\ 0, & \text{if label = clean.} \end{cases}$$

This approach treats severity as directly proportional to the model's certainty in detecting dust [7]. In cases where dust is absent, severity defaults to zero. For borderline predictions (e.g., partial dust coverage or ambiguous patterns), scaled confidence values provide a smooth, interpretable gradient of severity rather than binary outcomes [21]. A fallback mechanism is incorporated to handle anomalous or undefined model outputs, thereby ensuring stability and consistency in severity reporting [6].

(a) Dust defect


(b) Dust Prediction

Fig. 6: Dust detection pipeline

*4) **Hotspot Detection using YOLOv8-Thermal**:* The hotspot detection module was implemented using the YOLOv8 thermal object detection framework [12], [29], [31], selected for its strong balance between detection accuracy and real-time operability. As a single-stage detector, YOLOv8 offers computational efficiency and consistent anomaly localization, making it well-suited for drone-based photovoltaic (PV) inspection where rapid processing is essential [30].

The architecture integrates a lightweight feature-extraction backbone, a multi-scale feature aggregation neck, and a unified detection head enabling joint execution of class prediction, bounding box regression, and objectness estimation. This combined design effectively captures localized temperature deviations while maintaining low inference latency, ensuring reliable detection performance in operational environments [11].

The model was fine-tuned on a curated dataset of thermal solar panel images representing both normal and anomalous thermal signatures [4]. Training was performed using the YOLOv8n configuration with an input resolution of $640 \times 640$ pixels, a batch size of 16, and a total of 30 epochs. The Stochastic Gradient Descent (SGD) optimizer with warm-up scheduling was employed to ensure stable convergence [19].

Model performance was evaluated using standard COCO metrics, including Precision, Recall, and mAP@0.5 [31]. During inference, the detector achieved an average processing speed of approximately 7–8 ms per frame, demonstrating its suitability for real-time deployment [29].

**Unified Severity Scoring Model:** Hotspot severity was determined using a composite formulation integrating predicted anomaly type, prediction confidence, and spatial extent. Each hotspot class was associated with a normalized severity weight reflecting its relative operational criticality. Table I shows the category-to-severity mapping [30].

TABLE I: Severity weights for hotspot anomaly classes

| **Class ID** | **Severity Weight $W_{\text{class}}$** | **Description** |
|---|---|---|
| 0 | 0.20 | Single Hotspot |
| 1 | 0.40 | Multi Hotspots |
| 2 | 0.30 | Single Diode Anomaly |
| 3 | 0.60 | Multi Diode Anomalies |
| 4 | 0.50 | Single Bypassed Substring |
| 5 | 0.70 | Multi Bypassed Substring |
| 6 | 0.85 | String-level (Open Circuit) |
| 7 | 1.00 | String-level (Reversed Polarity) |

To incorporate spatial extent, the bounding-box area is normalized with respect to the overall image resolution:

$$\text{area\_factor} = \frac{(x_2 - x_1)(y_2 - y_1)}{W_{\text{img}} H_{\text{img}}},$$

where $(x_1, y_1, x_2, y_2)$ denote bounding-box coordinates and $W_{\text{img}}, H_{\text{img}}$ represent image width and height, respectively.

For each detected hotspot, the bounding-box severity score is computed as:

$$S_{\text{box}} = W_{\text{class}} \times \text{conf} \times (1 + \text{area\_factor}),$$

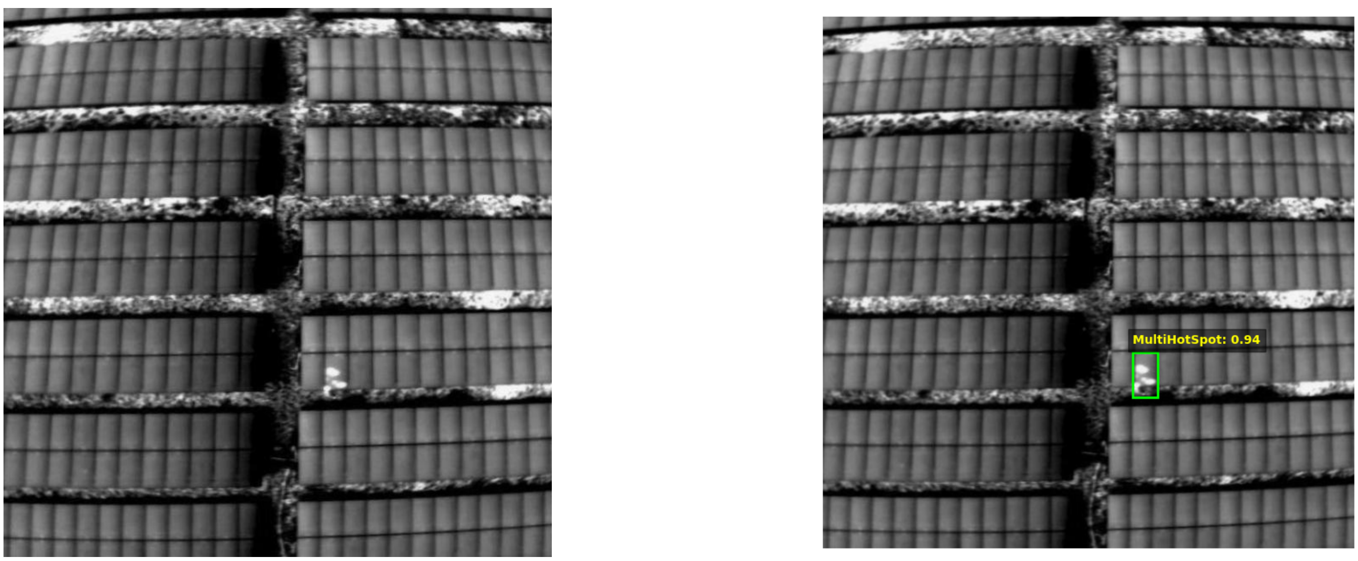
where $W_{\text{class}}$ is the severity weight, conf is the YOLOv8 prediction confidence score, and area_factor represents the normalized area contribution. The multiplicative confidence term ensures that lower-certainty predictions contribute proportionally less to the overall severity metric [29], [31].

The image-level severity score is then computed as:

$$S_{\text{image}} = \frac{1}{N} \sum_{i=1}^{N} S_{\text{box},i},$$

where $N$ is the number of detected hotspots in the image. This formulation enables severity estimation that jointly reflects anomaly type, confidence-weighted detection certainty, and spatial impact.

The YOLOv8-based detector demonstrated robust and reproducible hotspot identification across a broad range of thermal profiles and anomaly classes [12], [30]. Its combination of rapid inference and strong detection accuracy highlights its suitability for PV inspection tasks. In comparative evaluations, YOLOv8 outperformed heavier two-stage architectures such as Faster R-CNN in both latency and consistency of thermal defect detection [31], underscoring its effectiveness for real-time monitoring applications.




(a) Hotspot defect (b) Hotspot Prediction

Fig. 7: Hotspot detection pipeline

*5) **PID Severity Estimation using $\Delta T$ and Spatial Clustering**:* The PID assessment module integrates thermal anomaly localization, $\Delta T$-based temperature quantification, and spatial clustering into a unified severity estimation model. The pipeline operates on YOLO-formatted annotations generated from the hotspot detection stage [29]–[31]. Each annotation follows the standard YOLO structure $(\text{class\_id}, x_c, y_c, w, h)$, where $x_c$ and $y_c$ denote the normalized bounding-box center coordinates, and $w$ and $h$ represent the normalized width and height.

These parameters are denormalized into absolute pixel coordinates as:

$$x_{\min} = (x_c - \tfrac{w}{2})W, \qquad y_{\min} = (y_c - \tfrac{h}{2})H,$$
$$x_{\max} = (x_c + \tfrac{w}{2})W, \qquad y_{\max} = (y_c + \tfrac{h}{2})H,$$

where $W$ and $H$ denote the image width and height, respectively. This reconstruction enables precise extraction of thermal regions of interest corresponding to PID-affected cells [11], [12], [14].

**Based Thermal Stress Estimation:** For each annotated region, pixel-level thermal contrast analysis is performed to quantify thermal stress attributable to PID. The relative temperature elevation is computed as:

$$\Delta T = T_{\text{max, anomaly}} - T_{\text{avg, reference}},$$

where $T_{\text{max, anomaly}}$ is the maximum grayscale intensity within the bounding box, and $T_{\text{avg, reference}}$ denotes the mean intensity of thermally stable background regions [29].

The maximum observed $\Delta T$ is normalized against a predefined threshold $(\Delta T_{\text{MAX}} = 50°C)$ [32]:

$$\Delta T_{\text{norm}} = \min\left(\frac{\Delta T_{\max}}{\Delta T_{\text{MAX}}},\ 1.0\right),$$

yielding a bounded thermal stress indicator robust to environmental variations [33], [34].

**Spatial Clustering of PID Anomalies:** PID degradation typically manifests as spatially correlated thermal anomalies across electrically coupled cells [32], [34]. To capture this behavior, centroid coordinates of all bounding boxes are computed as:

$$C_i = \left(\frac{x_{\min} + x_{\max}}{2},\ \frac{y_{\min} + y_{\max}}{2}\right).$$

Centroids are clustered using DBSCAN with parameters $\varepsilon = 60$ pixels and MinPts $= 1$. DBSCAN identifies dense anomaly regions while allowing detection of isolated hotspots [31].

The PID propagation index is defined as:

$$R_{\text{PID}} = \frac{\text{size of largest cluster}}{\text{total hotspots}},$$

where values approaching unity indicate highly concentrated anomaly clusters characteristic of severe PID spread across electrically continuous substrings [30], [32].

To quantify anomaly dispersion, the ratio of detected clusters to total hotspots is computed as:

$$D = \min\left(\frac{k}{n},\ 1.0\right),$$

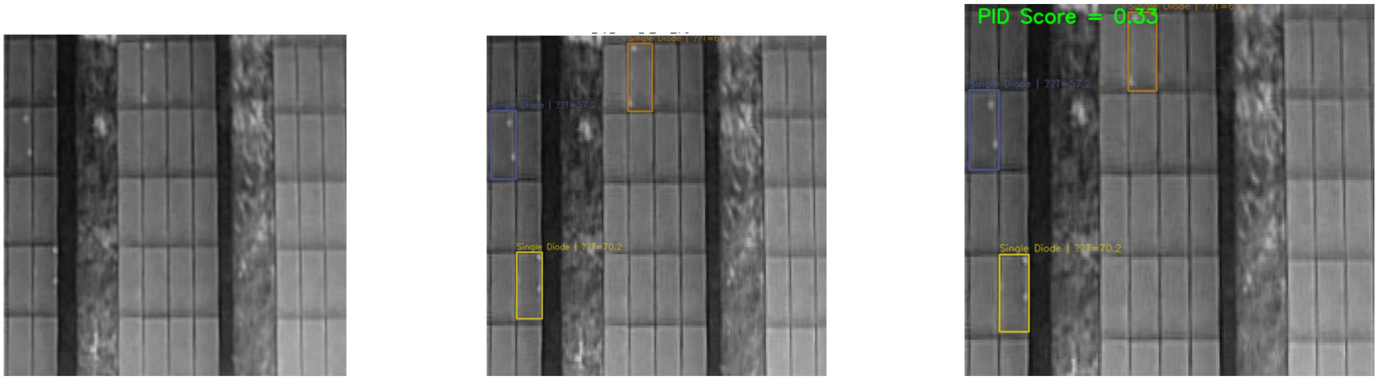
where $k$ is the number of clusters and $n$ is the total number of hotspots. This term penalizes spatially scattered defects that are less indicative of PID propagation [33].

**Unified Severity Scoring Model:** The final PID severity score integrates thermal stress, propagation density, and cluster dispersion using a weighted aggregation model [34]:

$$S = w_{\Delta}\, \Delta T_{\text{norm}} + w_{\text{PID}}\, R_{\text{PID}} + w_{\text{cluster}}\, D,$$

where weights are empirically selected as $w_{\Delta} = 0.5$, $w_{\text{PID}} = 0.4$, and $w_{\text{cluster}} = 0.1$. These values assign primary importance to thermal stress, secondary importance to spatial concentration of anomalies, and supplemental significance to anomaly dispersion.

This integrated model captures both intensity-driven and topology-driven degradation characteristics, providing a robust, temperature-aware indicator of PID severity suitable for automated solar module health diagnostics [29], [32], [34].




(a) PID defect (b) PID preprocessing (c) PID prediction

Fig. 8: Discoloration detection pipeline

### *C. Degradation Scoring*

The Impact Weight Assignment converts raw defect detections into weighted severity values that quantify the operational impact of each fault on photovoltaic (PV) module performance. The weighting mechanism is derived from peer-reviewed industry literature [38]–[40], IEEE PV reliability standards, and state-level solar guidelines such as the Tamil Nadu Solar PV Policy and associated technical whitepapers.

Tamil Nadu installations are often subjected to semi-arid, coastal, and high-dust environments, resulting in high soiling frequency [36], [37]. Cracks and hotspots drive urgent maintenance actions, while discoloration manifests gradually. PID is significant but less prevalent in visual datasets, justifying its lower relative weighting [40].

Each defect type exhibits a distinct contribution to long-term degradation and short-term performance loss; therefore, weight calculations are tailored to the defect's physical characteristics, failure propagation behaviour, and empirical impact levels reported in the literature [38]. All assigned weights are normalized to a continuous range of $0$–$1$, ensuring a uniform scale for multi-defect comparison and enabling direct aggregation into a unified PV degradation score.

TABLE II: Proposed Impact Weightings for Defect Severity Contribution

| **Defect Type** | **Weight (%)** |
|---|---|
| Dust / Soiling | 30% |
| Cracks (micro/macro) | 25% |
| Hotspots | 20% |
| Discoloration (EVA, delamination) | 15% |
| Potential-Induced Degradation (PID) | 10% |

- *Dust (30%):* Multiple India-based studies identify soiling losses as a major contributor to PV output reduction, with Tamil Nadu reporting an approximate annual soiling-related loss of $\sim 6.5\%$ [36], [37]. The relatively high frequency of dust accumulation in semi-arid and industrial regions justifies assigning the largest weight.

- *Cracks (25%):* Cracks, both micro and macro, are consistently identified in PV reliability literature as critical degradation pathways that lead to cell isolation, mismatch losses, and long-term structural damage [38]. Their high maintenance significance supports a substantial weight.
- *Hotspots (20%):* Hotspots represent thermal concentration zones that accelerate local degradation and pose electrical safety risks. Although regional studies are limited, global reliability reviews place hotspots as high-risk failure indicators, motivating a moderate-high weight [39].
- *Discoloration (15%):* Encapsulant discoloration and EVA degradation are ageing-driven defects that impact optical transmission efficiency. While significant, they generally progress slowly relative to cracks and soiling, justifying a moderate weight [38].
- *Potential-Induced Degradation (PID) (10%):* PID is a known degradation mode influenced by system configuration, high voltages, humidity levels, and module grounding conditions. However, its detectability via imagery alone is limited, and its overall incidence frequency is lower. Consequently, a conservative weight is used [40].

***Final PV Module Degradation Score:*** For a module with $M = 5$ defect categories, the overall degradation score $S_{\text{deg}}$ is computed as a weighted sum of the severity values output by each defect-specific module:

$$S_{\text{deg}} = \sum_{i=1}^{M} (S_i \times W_i),$$

where:

- $S_i$ is the normalized severity score for defect type $i$,
- $W_i$ is the corresponding impact weight from Table II,

This aggregation yields a scalar degradation score in the range $[0, 1]$, interpretable as the normalized health-impact level of the PV module.

## IV. Experimental Setup

### A. *Hardware Configuration*

The experiments were conducted on a workstation equipped with an NVIDIA Tesla T4 GPU (16 GB VRAM), Intel Xeon CPU @ 2.30 GHz, and 32 GB RAM. This setup ensured efficient training and inference for computationally intensive deep learning architectures. All experiments were performed on Google Colab, leveraging its stable cloud-based CUDA environment.

### B. *Software Frameworks*

The implementation was carried out in Python 3.10, using PyTorch 2.2.0 for model training and TensorFlow 2.15 for fine-tuning the Vision Transformer (ViT) classifier. Supporting libraries included OpenCV for image processing, NumPy and Matplotlib for data manipulation and visualization, and Scikit-learn for performance evaluation. Dataset annotations and augmentations were created using Roboflow to ensure consistent preprocessing across both RGB and thermal datasets.

### C. *Training Hyperparameters*

All models were trained using their default Roboflow-provided hyperparameters unless otherwise specified. The Adam optimizers were employed depending on the model implementation, while learning rates, batch sizes, and scheduling parameters followed the respective official configurations. Standard data augmentation strategies, including random rotation, horizontal flipping, and brightness normalization, were applied to improve robustness against illumination changes, dust accumulation, and panel orientation variations.

For YOLO-based detectors (YOLOv8/YOLOv11 variants), the models were trained to detect multiple PV defects such as cracks, discoloration, hotspots, and potential-induced degradation (PID). Default anchor-free detection settings and mosaic-based augmentation were retained to stabilize convergence across heterogeneous panel conditions.

Mask R-CNN models were also trained for pixel-level segmentation of cracks and discoloration. For crack segmentation, the Detectron2 implementation with a ResNet-101-FPN backbone pretrained on COCO was fine-tuned using a mini-batch size of 2, a base learning rate of $5 \times 10^{-4}$, a maximum of 8000 iterations, and a warmup-free LR schedule. Input resolution was fixed at 640×640 for both training and validation, while ROI head batch size was set to 512 to improve positive sample representation.

A second Mask R-CNN variant was used for hotspot and thermal defect detection. The model was trained using a ResNet-50-FPN backbone with AdamW optimization (learning rate $1 \times 10^{-4}$) for 5 epochs. COCO-format annotations were parsed through custom loading functions, bounding boxes were converted to $(x_1, y_1, x_2, y_2)$ format, and batches containing at least one valid object instance were retained. The model was fine-tuned for 8 defect classes plus background, and the checkpoint was saved for downstream evaluation on thermal imagery.

### D. *Evaluation Metrics*

Model performance was evaluated using four standard metrics: Precision (P), Recall (R), F1-Score ($F_1$), and Mean Average Precision (mAP). These metrics collectively assess both detection accuracy and robustness [35].

*1) Precision:* Precision quantifies the proportion of correctly identified defective regions among all detected ones:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP and FP denote true positives and false positives, respectively.

*2) Recall:* Recall measures the proportion of actual defective samples correctly detected:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN represents false negatives.

*3) F1-Score:* The F1-Score provides the harmonic mean between Precision and Recall:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

*4) Mean Average Precision (mAP):* Mean Average Precision is defined as:

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i$$

where $AP_i$ represents the average precision for class $i$, and $N$ is the total number of defect classes.

*5) Average Precision at IoU = 0.75 (AP75):* AP75 is a stricter variant of average precision computed at a fixed Intersection-over-Union (IoU) threshold of $0.75$:

$$AP_{75} = AP(\text{IoU} = 0.75)$$

A higher AP75 value indicates that the detected bounding boxes or segmentation masks align more precisely with the ground truth, implying strong localization accuracy. Since IoU $0.75$ requires significantly tighter overlap compared to IoU $0.5$, AP75 is often considered a robustness measure for fine-grained defect localization and shape conformity.

This setup ensures reproducibility, scalability, and reliability in solar panel defect detection. The combination of robust architectures, consistent training conditions, and standardized evaluation metrics enables accurate and efficient solar module health assessment.

## V. Results and Discussion

TABLE III: Performance comparison of different models across defect types.

| Defect Type | Model | mAP@0.5 | AP75 | Precision (%) | Recall (%) |
|---|---|---|---|---|---|
| Cracks | YOLOv11 | 65.4 | – | 100 | 51.4 |
| | Roboflow IS 3.0 | 35.2 | – | 48.2 | 31.0 |
| | RCNN | 15.63 | 11.83 | – | – |
| Hotspot | YOLOv8 | 93.8 | 84.7 | 83.6 | – |
| | RCNN | 64.01 | 50.57 | – | – |
| Discoloration | YOLOv11 | 37.5 | – | 81.7 | 33.2 |
| | Roboflow IS 3.0 | 82.3 | – | 84.6 | 78.6 |
| | RCNN | 24.33 | 5.47 | – | – |
| Dust | ViT-based classifier | 93.7 (Accuracy) | – | classification-based | classification-based |
| PID | Hotspot model + clustering logic | Inherits YOLOv8 | Inherits YOLOv8 | Inherits | Inherits |

**Table III** presents the performance of the evaluated models across cracks, hotspots, discoloration, dust, and PID estimation. The results directly reflect the defect-specific challenges described earlier.

*Crack Detection:* For cracks, YOLOv11 achieves the highest mAP@0.5 (65.4) and perfect precision (100%), but recall remains low (51.4). This indicates that while the model rarely misclassifies non-crack regions, it consistently fails to capture fine or micro-cracks. As noted, cracks are thin, irregular, and often blend with module texture and shadows, leading to missed detections. Roboflow Instance Segmentation 3.0 obtains lower accuracy (35.2 mAP), consistent with its difficulty distinguishing between true crack boundaries and panel edges under uneven lighting. RCNN performs the poorest (15.63 mAP, AP75 11.83), confirming its inability to generalize to small-scale, low-contrast defects due to weak region proposals and sensitivity to dataset size.

*Hotspot Detection:* Hotspot detection shows strong performance for YOLOv9 (mAP 93.8, AP75 84.7), aligned with the high thermal contrast of hotspots. The model reliably identifies bright IR signatures with minimal false positives. RCNN achieves lower accuracy (64.01 mAP, AP75 50.57), reflecting difficulty capturing diffuse or blurred hotspot boundaries typical of thermal imagery. Its bounding box predictions often shift or fail under low temperature gradients, as described.

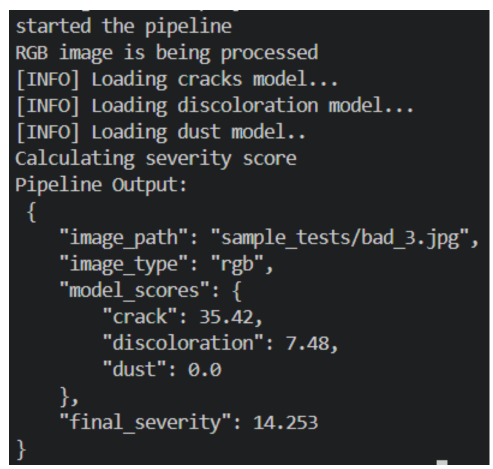
*Discoloration Detection:* For discoloration, YOLOv11 underperforms (37.5 mAP, recall 33.2), consistent with its reliance on well-defined object edges. Discoloration involves gradual tone variation, which does not form sharply bounded regions. Roboflow Instance Segmentation 3.0 performs best (82.3 mAP), directly reflecting its strength in pixel-level differentiation of texture and tonal shifts. RCNN again shows weak performance (24.33 mAP, AP75 5.47), matching its tendency to miss irregular and diffuse patterns.

*Dust Classification:* Dust detection is treated as a classification task using a ViT-based classifier, hence not reported under mAP-based metrics. Its outputs depend on classification performance rather than bounding box localization, which aligns with dust being spread across large regions rather than forming discrete objects.

*PID Estimation:* PID results inherit the detection performance of YOLOv9, as the PID module relies on hotspot outputs combined with clustering and $\Delta T$-based logic. Since PID is not detected directly but inferred from hotspot characteristics, its accuracy is dependent on YOLOv9's reliability.

*Multi-Defect Prediction Capability :* A key novelty of the proposed system lies in its ability to process a single input image and predict multiple defect types simultaneously. Rather than relying on one model to generalize across all defect characteristics, the pipeline integrates specialized components based on the strengths identified in Table X: YOLO models for structural and thermal anomalies, segmentation for tonal variations, and classification for diffuse dust patterns. This modular approach enables reliable detection of co-existing defects—such as simultaneous presence of cracks, discoloration, dust, PID and hotspots, which can occur in real photovoltaic modules. The unified VS Code pipeline operationalizes these models sequentially, generating consolidated outputs for all defect categories, from a single inference run. This multi-defect prediction capability directly addresses a gap in existing approaches that typically treat defects in isolation, establishing the system's primary contribution and novelty.

(a) Image with Crack + Discoloration




(b) Multi-defect Prediction

Fig. 9: Multi defect detection pipeline

## VI. Conclusion

**Challenges:** This variation in the performance of the models for different defect categories can be attributed to various factors operating together, including quality of data, variability in the environment, and deep learning model architecture constraints. [11], [18], [38].

While the quality of the dataset itself inherently defines the behavior, inconsistent lighting conditions, image resolutions, and varied angles make the representations of solar panel defects inconsistent. This makes the models fail to classify or recognize subtle patterns, especially when the defect consists of fine cracks or minor discoloration [24], [25]. Furthermore, class imbalance biases the learning toward the dominant classes of defects, hence weakening generalization for other classes. This is immediately reflected in recall: reduced sensitivity in detecting micro-cracks and faint discolorations [37].

Another challenge arises from annotation errors introduced in the process of manual labeling. Ambiguity in delineating the defect boundaries—especially for faint defects such as micro-cracks or early discoloration— introduces noise into the training data, reducing model precision [7]. Environmental disturbances such as dust accumulation, surface reflections, glare, and partial shading further obscure the true defect boundaries, leading to false positives and negatives during detection [2].

Each of these model architectures is differently sensitive to this kind of visual complexity. Region-based CNNs, like R-CNN, rely on coarse region proposals and hence may fail to capture key fine structural patterns. At the other extreme, YOLO-based detectors optimized for real-time performance often emphasize object-level features at the expense of finer textures. In contrast, pixel-level instance segmentation frameworks like Roboflow Instance Segmentation 3.0 are highly susceptible to inconsistencies in annotation and changes in lighting conditions. Besides these issues, insufficient preprocessing-including a lack of contrast normalization or histogram equalization in both RGB and thermal imagery-can mask weak yet informative signals that are important for accurate classification of low-intensity defects.

Taken collectively, these challenges have underlined the need for a balance in datasets, improved protocols for annotation, and domain-specific preprocessing pipelines that will enhance model robustness, interpretability, and transferability across diverse environmental conditions and defect types.

**Future Work:** The extensions in this work will resolve these limitations and widen the scope of solar panel defect analysis. Primary research directions involve the integration of RGB and thermal modalities within a single deep learning framework for enhanced cross-domain consistency and accuracy [], and multi-modal fusion for distinguishing between defect patterns that may be overlapping, thus improving interpretability by combining visible-spectrum and temperature-based cues.

Another promising avenue for the deployment of trained models on drones and on edge devices would be to enable real-time monitoring of large-scale solar farms [42], at which time techniques such as model compression, quantization, and pruning will ensure efficient on-device inference. Future research shall be performed on predictive maintenance and lifetime forecasting by analyzing temporal trends in defect progression using recurrent or transformer-based architectures [43].

In order to enhance the robustness of the model, the future work will expand the datasets by introducing advanced augmentation and synthetic generation techniques such as generative adversarial networks and diffusion-based models, which will simulate various lighting, angular, and weather conditions [44]. Model interpretability will also be enhanced by applying explainable AI methods to enhance the transparency of severity scoring for strengthening operator trust [43]. Besides, severity calibration will shift from heuristic weighting to data-driven optimization emanating from the correlations between the detected defects and actual power degradation [33]. All these put together will transform the proposed system from a detection-focused model to an intelligent, comprehensive platform for predictive and real-time solar asset management.

**Summary:** This research has successfully developed a multi-modal solar panel defect detection and severity assessment system using specialized deep learning architectures to analyze both RGB and thermal imagery. The work has targeted the critical challenge of automatic visual fault detection in photovoltaic modules and has demonstrated that modular defect-specific model integrations enhance the reliability in detection across a wide range of fault categories.

The system's end-to-end automated pipeline identifies input type - either RGB or thermal-, routes the input to the appropriate detection module, and computes per-defect severity and a final degradation score between 0 and 1. For RGB inputs, YOLOv11 achieved 65.4% mAP@0.5 in crack detection, while Roboflow Instance Segmentation 3.0 yielded 82.3% mAP@0.5 for discoloration [16]. Dust detection using a Vision Transformer (ViT) model attained an accuracy of 93.7%, and YOLOv9 achieved 93.8% mAP@0.5 with recall above 80% for hotspot detection [29], [39]. Effective PID severity inference through hotspot clustering introduced a new contribution into thermal analysis [33], [40] The most important contribution of this work is the associated severity scoring mechanism that translates the detection outputs into interpretable maintenance metrics. For each defect, a severity measure was calculated based on quantifiable parameters: crack length, discoloration area, hotspot spread, and dust confidence levels, all aligned with regional solar maintenance standards such as the Tamil Nadu Solar Policy [8]. This yields a unified health indicator, given by this degradation index, which facilitates informed decisions on preventive maintenance. While dataset imbalance, lighting sensitivity, and heuristic-based weighting remain limitations, the developed architecture establishes a strong foundation for scalable, field-ready solar inspection systems. Overall, this research demonstrates that tailored deep learning architectures combined with multi-modal analysis can deliver robust, interpretable, and automated assessments of solar panel degradation [41], [43]. By bridging the gap between visual inspection and actionable maintenance analytics, this system contributes a key step toward sustainable, data-driven solar asset management. [42], [44].

## References


[1] The Times of India, "A routine storm reveals the high price of poor quality solar," *The Times of India*, Mar. 2, 2025. [Online]. Available: https://timesofindia.indiatimes.com/business/india-business/a-routine-storm-reveals-the-high-price-of-poor-quality-solar/articleshow/120999924.cms

[2] V. Subha, C. T. Lakshmi, and S. Subbulakshmi, "A Study on Barriers for Implementing Solar Energy Initiatives in Tirunelveli and Thoothukudi District," *NeuroQuantology*, vol. 20, no. 19, pp. 5930–5938, Nov. 2022, doi: 10.48047/nq.2022.20.19.nq99583.

[3] M. Afroz, "Solar Panel Images," Kaggle, 2022. [Online]. Available: https://www.kaggle.com/datasets/pythonafroz/solar-panel-images

[4] P. K. Darabi, "Solar Panel Thermal Images," Kaggle, 2021. [Online]. Available: https://www.kaggle.com/datasets/pkdarabi/solarpanel

[5] "Investigation of Degradation of Solar Photovoltaics: A Review of Aging Factors, Impacts, and Future Directions Toward Sustainable Energy Management," *Energies*, vol. 16, no. 21, pp. 1–22, 2023. doi: 10.3390/en16218345.

[6] R. Nair and S. Menon, "Analysis of Dust Accumulation Effects on Long-Term Performance of Solar PV Panels," *AIMS Energy*, vol. 13, no. 2, pp. 221–236, 2025. doi: 10.3934/energy.2025012.

[7] S. Prakash and V. Kumar, "Dust Detection in Solar Panels Using Image Processing Techniques: A Review," *Research, Society and Development (RSD)*, vol. 9, no. 11, pp. 1–10, 2020. doi: 10.33448/rsd-v9i11.10092.

[8] Tamil Nadu Energy Department, *Annual Energy Report 2024–25*. Government of Tamil Nadu, 2024. [Online]. Available: https://www.tn.gov.in/dept/energy

[9] J. Hunt, M. Sharma, and K. Pillai, "Quantifying Renewable Energy Potential and Capacity in India," *Renewable Energy Reports*, vol. 12, pp. 55–68, 2024. doi: 10.1016/j.rer.2024.03.005.

[10] S. M. Kumar, "Scenario of Solar Energy and Policies in India," *International Journal of Energy Policy Studies*, vol. 8, no. 1, pp. 10–19, 2024. doi: 10.1016/j.ijeps.2024.01.002.

[11] S. R. Mohanty, "Machine Learning Approaches for Automatic Defect Detection in Solar Panels," *Solar Energy*, vol. 275, pp. 113672–113682, 2025. doi: 10.1016/j.solener.2025.113672.

[12] A. Patel et al., "Automated Fault Detection in PV Modules Using YOLOv8 and Deep Learning Frameworks," *Solar Energy Materials Solar Cells*, vol. 265, pp. 112487–112496, 2024.

[13] H. Kim and Y. Zhao, "YOLOv9 for Photovoltaic Defect Localization under Variable Illumination," *IEEE Access*, vol. 12, pp. 34456–34467, 2024.

[14] P. Singh et al., "Instance Segmentation of PV Cell Defects Using Mask R-CNN and DeepLab," *Applied Energy*, vol. 334, pp. 120080–120092, 2023.

[15] L. Rodrigues and F. Chen, "Semantic Segmentation for Microcrack Identification in Solar Cells," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 3, pp. 3555–3567, 2024.

[16] J. H. Lambert and R. Smith, "Roboflow: Streamlining Dataset Curation and Annotation for Vision Models," *arXiv preprint arXiv:2306.14512*, 2023.

[17] T. Verma and A. Deshpande, "Crack Propagation and Electrical Impact in Monocrystalline Silicon Solar Cells," *Solar Energy*, vol. 255, pp. 111412–111423, 2023.

[18] "Investigation of Degradation of Solar Photovoltaics: A Review of Aging Factors, Impacts, and Future Directions Toward Sustainable Energy Management," *Energies*, vol. 16, no. 21, pp. 1–22, 2023.

[19] G. Li et al., "Benchmarking Deep Learning Models for Solar Panel Defect Detection," *IEEE Access*, vol. 12, pp. 49877–49889, 2024.

[20] A. Shaik, A. Balasundaram, L. S. Kakarla, and N. Murugan, "Deep Learning-Based Detection and Segmentation of Damage in Solar Panels," *Automation*, vol. 5, no. 2, pp. 128–150, 2024. doi: 10.3390/automation5020008.

[21] J. Imaging, "Solar Panel Surface Defect and Dust Detection: Deep Learning Approach," *Journal of Imaging*, vol. 11, no. 9, Art. 287, 2025. doi: 10.3390/jimaging11090287.

[22] "Detection and Classification of Photovoltaic Module Defects Based on Artificial Intelligence," *Neural Computing and Applications*, vol. 36, pp. 16769–16796, 2024. doi: 10.1007/s00521-023-09325-8.

[23] R. Arora, S. Jain, and P. S. Mehta, "Transformer-Based Image Classification for Surface Contamination in Photovoltaic Modules," *Renewable Energy*, vol. 220, pp. 120–132, 2024. doi: 10.1016/j.renene.2023.09.044.

[24] K. Verma and M. Goyal, "Analysis of Discoloration and Soiling in PV Modules Using Image-Based Diagnostics," *Solar Energy Materials  Solar Cells*, vol. 256, pp. 111432–111441, 2023. doi: 10.1016/j.solmat.2023.111432.

[25] F. Rahman, A. Roy, and D. Biswas, "Discoloration in Silicon Solar Cells: Causes, Detection, and Mitigation Strategies," *IEEE Journal of Photovoltaics*, vol. 13, no. 4, pp. 765–774, 2023. doi: 10.1109/JPHOTOV.2023.3265410.

[26] L. Zhang, Y. Li, and J. Zhou, "Vision Transformers for Environmental Soiling Detection on PV Modules," *Applied Energy*, vol. 356, pp. 121567–121580, 2024. doi: 10.1016/j.apenergy.2024.121567.

[27] A. Alharbi and M. Rehman, "Image-Based Detection of Dust and Surface Anomalies in PV Panels Using Deep Learning," *Energy Reports*, vol. 9, pp. 11201–11212, 2023. doi: 10.1016/j.egyr.2023.09.042.

[28] S. Ahmad and B. Pradhan, "Deep Learning for Dust Severity Classification on Solar Modules Under Real-World Conditions," *IEEE Access*, vol. 12, pp. 54001–54015, 2024. doi: 10.1109/ACCESS.2024.3402125.

[29] D. Wang, P. Yan, C. Yao, B. Xiao, W. Zhao, and R. Zhu, "A lightweight joint metric detection approach on YOLO for hot spots in photovoltaic modules," *AIP Advances*, vol. 10, no. 6, Art. 065221, 2024. doi:10.1063/5.0232136.

[30] "Inspection Method Enabled by Lightweight Self-Attention for Multi-Fault Detection in Photovoltaic Modules," *Electronics*, vol. 14, no. 15, Art. 3019, 2024. doi:10.3390/electronics14153019.

[31] T. Verma and A. Deshpande, "Comparative Performance Evaluation of YOLOv5, YOLOv8, and YOLOv11 for Solar Panel Defect Detection," *Preprints.org*, 2025. doi:10.20944/preprints202501.0788.

[32] W. Luo, Y. S. Khoo, P. Hacke, V. Naumann, D. Lausch, S. P. Harvey, J. P. Singh, J. Chai, Y. Wang, A. G. Aberle and S. Ramakrishna, "Potential-induced degradation in photovoltaic modules: A critical review," *Energy  Environmental Science*, vol. 10, no. 1, pp. 43–68, 2017. doi:10.1039/C6EE02271E.

[33] C. Molto, J. Oh, F. I. Mahmood, M. Li, F. Li, R. Smith, D. Colvin, M. Matam, C. DiRubio and G. Tamizhmani, "Review of potential-induced degradation in bifacial photovoltaic modules," *Energy Technology*, 2022. doi:10.1002/ente.202200943.

[34] F. Ibne Mahmood, F. Li, P. Hacke, C. Molto, D. Colvin, H. Seigneur and G. Tamizhmani, "Susceptibility to polarization-type potential induced degradation in commercial bifacial p-PERC PV modules," *Progress in Photovoltaics*, 2023. doi:10.1002/pip.3724.

[35] T. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C. Zitnick, and P. Dollár, "Microsoft COCO: Common Objects in Context," *European Conference on Computer Vision (ECCV)*, 2014.

[36] P. Rajput, A. Singh, and S. Narayanan, "Soiling Losses and Cleaning Optimization for Solar PV Systems in India: A Regional Assessment," *Renewable Energy*, vol. 180, pp. 1025–1037, 2021. doi: 10.1016/j.renene.2021.08.020.

[37] R. Krishnan and V. Subramanian, "Inside the Impacts of Soiling on Solar Modules: Case Study from Vellore, Tamil Nadu," *PV Magazine International*, 2022. Available: https://www.pv-magazine.com/2022/07/15/inside-the-impacts-of-soiling/.

[38] A. Jangid, K. S. Reddy, and T. K. Mallick, "Comprehensive Review on Reliability and Failure Modes of Photovoltaic Modules," *Energies (MDPI)*, vol. 13, no. 16, pp. 4121–4142, 2020. doi: 10.3390/en13164121.

[39] Y. Wang, R. M. Behera, and L. A. Rosales, "Hotspot Defects in Photovoltaic Modules: Mechanisms, Detection, and Reliability Implications," *Progress in Photovoltaics: Research and Applications (OUP)*, vol. 29, no. 8, pp. 841–856, 2021. doi: 10.1002/pip.3415.

[40] K. Tanaka, M. Takashima, and T. Kato, "Potential-Induced Degradation in Photovoltaic Modules: Mechanisms, Impacts, and Mitigation," *Solar Energy Materials and Solar Cells*, vol. 200, pp. 109935, 2019. doi: 10.1016/j.solmat.2019.109935.

[41] T. Zhang, L. Chen, and Q. Zhao, "Multi-Modal Fusion of RGB and Infrared Images for Photovoltaic Fault Detection," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 5, pp. 5721–5733, 2024. doi: 10.1109/TII.2024.3456712.

[42] K. Singh and D. Patel, "Drone-Based Automated Inspection of Solar Farms Using Deep Learning," *Renewable and Sustainable Energy Reviews*, vol. 178, pp. 113200, 2023. doi: 10.1016/j.rser.2023.113200.

[43] P. Das and S. Roy, "Explainable AI for Solar Fault Detection: Enhancing Transparency and Trust in Deep Learning Models," *Energy AI*, vol. 15, pp. 100335, 2024. doi: 10.1016/j.egyai.2024.100335.

[44] Y. Liu and C. He, "Synthetic Data Generation for Photovoltaic Fault Detection Using GANs and Diffusion Models," *Energy Reports*, vol. 9, pp. 11925–11938, 2023. doi: 10.1016/j.egyr.2023.09.056.